Reishi Mitani

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SUMMARY

Backend and platform engineer focused on applied ML systems, data infrastructure, and production reliability.
Experienced in integrating machine-learning pipelines into large-scale services, improving observability for high-traffic APIs, and leading data-model and storage migrations.
Currently working at Indeed on taxonomy and metadata platforms supporting ML-driven features at scale.
Fluent in English and Japanese; professional proficiency in Chinese.

EXPERIENCE

Software Engineer — Indeed Technologies Japan

2023 – Present

Work on backend platforms and data infrastructure powering ML-driven metadata and taxonomy services.

ML platform integration

- Contributed to the migration from rule-based attribute extraction to model-based extraction using transformer models (MMBERT)
- Implemented backend logic to reconcile model outputs with legacy pipelines and ensure safe rollout
- Supported staged production deployment and helped improve extraction quality and coverage while maintaining API compatibility

Observability & reliability at scale

- Improved logging and monitoring for metadata APIs handling ~800M requests/day

- Led reductions in redundant error logging, lowering Datadog costs and improving signal-to-noise for incidents
- Designed monitoring thresholds and contributed to SLO/latency analysis workflows for on-call engineers

Data platform modernization

- Led design and migration of taxonomy snapshot artifacts to a structured JSON format
- Reduced artifact size from ~550MB to ~45MB, improving distribution, parsing speed, and debugging workflows
- Coordinated migrations across downstream services and external consumers

Graph database migration

- Led POC and core importer for migrating taxonomy graph data from Neo4j to NebulaGraph
- Implemented large-scale node/relationship import pipelines (~600k nodes, ~4M edges)
- Contributed to schema mapping decisions and validation workflows to ensure data parity during dual-write rollout

Tech stack: Java, Kotlin, Go, MongoDB, Neo4j, AWS, gRPC, Datadog

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IoT Security Engineer (Part-time)

2022 – Present

Work on embedded security and device identity systems.

- Integrated Sony Spresense and STM32 boards into secure device pipelines
- Implemented decentralized-identity-based device authentication flows
- Built communication and cryptographic handling logic for constrained devices

Tech: C, Embedded Systems, Cryptography, DID, IoT Security

EDUCATION

M.S. Interdisciplinary Information Studies
University of Tokyo, 2021 Apr. - 2023 Mar.

B.Eng. Systems Innovation
University of Tokyo, 2016 Apr. - 2021 Mar.

SKILLS

Languages: Java, Kotlin, Go, TypeScript, C, Python
Infra/Data: MongoDB, Neo4j, AWS, distributed data pipelines
Specialties: ML integration, backend platforms, observability, system migrations, data modeling

Languages spoken:
English (fluent, TOEFL iBT 117)
Japanese (native)
Chinese (advanced, HSK6)